

Number system

① Integers $\rightarrow \{\dots -3, -2, -1, 0, 1, 2, \dots\}$

② Negative Integer $\rightarrow \{\dots -7, -6, -5, \dots -1\}$

③ Positive Integer $\rightarrow \{\cancel{0}, 1, 2, 3, \dots\}$

$[0 = \text{neutral}]$

∴ (4) Natural number → {1, 2, 3, ...}

or

Positive Integer

(5) Whole number → {0, 1, 2, 3, ...}

⑥ Imaginary No.

$$\sqrt{9} = 3$$

$$\sqrt{1} = 1$$

$$\sqrt{-1} = i = \text{iota}$$

$$i = \sqrt{-1}$$

$$i^2 = (\sqrt{-1})^2 = -1$$

$$i^3 = i^2 \cdot i = (-1)(i) = -i$$

$$i^4 = i^2 \cdot i^2 = (-1)(-1) = 1$$

$$\Rightarrow i + i^2 + i^3 + i^4 = 0$$

.. 8]

$$\frac{i^{12} + i^{13} + i^{14} + i^{15}}{i^{18} + i^{19} + i^{20} + i^{21}}$$

solⁿ :

$$\frac{i^{12} [1 + i + i^2 + i^3]}{i^{18} [1 + i + i^2 + i^3]}$$

WRONG

$$= i^{12-18} = i^{-6} = \frac{1}{i^6} = \frac{1}{i^4 \cdot i^2} = \textcircled{-1/1}$$

$$\Rightarrow \frac{i^{12} \left[1 + i + i^2 + i^3 \right]^0}{i^{18} \left[1 + i + i^2 + i^3 \right]^0}$$

$$= \frac{0}{0}$$

= indeterminate //

$$\Rightarrow [1 + i + i^2 + i^3]$$

$$= [\cancel{1} + i - \cancel{1} - i]$$

$$= 0$$

$$\textcircled{1} = \sqrt{1} = \sqrt{(-1)(-1)}$$

$$\sqrt{ab} \neq \sqrt{a} \sqrt{b}$$

$$\neq \sqrt{-1} \cdot \sqrt{-1}$$

$$\neq i \cdot i$$

$$\neq i^2$$

$$\neq \textcircled{-1}$$

$$\sqrt{ab} = \sqrt{a} \sqrt{b}$$



(i) $a = +, b = +$

(ii) $a = -, b = +$

(iii) $a = +, b = -$

(iv) $a = -, b = -$

$$\{ \sqrt{ab} \neq \sqrt{a} \sqrt{b} \}$$

⑦ Rational Number $[\frac{p}{q}]$

(1) $\frac{2}{3} = 0.6666 \dots$ {Recurring}

(2) $\frac{3}{2} = 1.5$ {finite}

(3) $\frac{3}{1} = 3$

⑧ Irrational number

Non-terminating

Non-Recurring

Ex : $\sqrt{3}$, $\sqrt{5}$

$$e = 2.7182818285 \dots$$

$$\pi = 3.1415926 \dots$$

(9) Proper fraction

$$\frac{p}{q}$$

$$-1 < \frac{p}{q} < 1$$

Ex : $\frac{1}{8}$

∴ (9) Im Proper fraction

$$\frac{p}{q}$$



$$\frac{p}{q} > 1, \quad \frac{p}{q} < -1$$

$$\frac{3}{2}$$

Mixed fraction [modified version of Improper fraction]

$$3\frac{1}{4} = \frac{(3 \times 4) + 1}{4} = \frac{13}{4}$$

⑩ Prime Number

2, 3, 5, 7, 11, 13, 17, 19 -

⑪ Composite No

4, 6, 8, 9 -

$\Rightarrow 1 = \text{Neither Prime}$
 Nor Composite

∴ Q] 341 . Is it prime or not?

$$\textcircled{1} \quad \sqrt{341} = 18.4 \approx \underline{19}$$

~~②~~ 2, 3, 5, 7, 11, 13, 17, 19

$$\textcircled{3} \quad \frac{341}{\text{Prime No.}} \Rightarrow \frac{341}{11} \rightarrow \text{It is divisible}$$

{ Not Prime }

Q] 179 Is it prime?

① $\sqrt{179} = 13.3 \approx 14$

② 2, 3, 5, 7, 11, 13, 17

③ $\frac{179}{\text{Prime No.}}$ → It is not divisible

⇒ It is prime //

Indices

$$\textcircled{1} \quad \underbrace{a \times a \times a \dots \times a}_{m \text{ times}} = a^m$$

$$\underbrace{a + a + a + \dots + a}_{m \text{ times}} = ma$$

∴

$$\textcircled{2} \quad a^m \times a^n = a^{m+n}$$

$$\textcircled{3} \quad a^{-1} = \frac{1}{a}$$

$$a^{-m} = \frac{1}{a^m}$$

$$\textcircled{4} \quad (a^m)^n = (a^n)^m = a^{m \times n}$$

∴ ⑤ $a^0 = 1 \quad \{a \neq 0\}$

⑥ $(ab)^m = a^m \cdot b^m$

⑦ $a^{m/n} = \sqrt[n]{a^m}$

⑧ $a^m = a^n \Rightarrow m = n$

8] $4^x \cdot 2^y = 128$

$$3^{3x} \cdot 3^{2y} - 9^{xy} = 0$$

$$x + y = ?$$

Solⁿ :

$$\textcircled{1} \quad (2^2)^x \cdot 2^y = 2^7$$

$$\Rightarrow 2^{2x} \cdot 2^y = 2^7$$

$$\Rightarrow 2^{2x+y} = 2^7$$

$$\boxed{2x + y = 7} \rightsquigarrow y = 7 - 2x$$

$$\underline{3^{3x}} \cdot \overbrace{3^{2y}} - 9^{xy} = 0$$

$$3^{3x+2y} = 9^{xy} = (3^2)^{xy} = 3^{2xy}$$

$$\boxed{3x + 2y = 2xy}$$

$$\Rightarrow 3x + 2(7 - 2x) = 2x(7 - 2x)$$

$$4x^2 - 15x + 14 = 0$$

$$a = 4$$

$$b = -15$$

$$c = 14$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = 2, 7/4$$

$$\Rightarrow y = 7 - 2x$$

$$\Rightarrow y = 3, 7/2$$

.

$$x + y = 5$$

Shrenik Jain — Study Simplified

Surds

$\sqrt{a}$ = irrational no.

$\sqrt{2}, \sqrt{3}, \sqrt{5} \dots$

Q] $x = \sqrt{3} + \sqrt{2}$

find $x^3 + \underline{x} + \frac{1}{\underline{x}} + \frac{1}{x^3}$ is ?

Rationalization
Concept

Soln : ① $x = \sqrt{3} + \sqrt{2}$

$$\frac{1}{x} = \frac{1}{(\underline{\sqrt{3}} + \underline{\sqrt{2}})} \cdot \frac{(\underline{\sqrt{3}} - \underline{\sqrt{2}})}{(\underline{\sqrt{3}} - \underline{\sqrt{2}})}$$

$$= \frac{\sqrt{3} - \sqrt{2}}{1} = \sqrt{3} - \sqrt{2}$$

$$(a-b)(a+b) = a^2 - b^2$$

$$\Rightarrow \left(x + \frac{1}{x}\right) + \left(x^3 + \frac{1}{x^3}\right)$$

$$\# (a+b)^3 = a^3 + b^3 + 3ab(a+b)$$

$$\Rightarrow \underline{a^3 + b^3 = (a+b)^3 - 3ab(a+b)}$$

$$g = \left(x + \frac{1}{x}\right)^3 - 3x \cdot \frac{1}{x} \left(x + \frac{1}{x}\right) + x + \frac{1}{x}$$

$$\therefore g = \left(x + \frac{1}{x}\right)^3 - 3 \cancel{x} \cdot \cancel{\frac{1}{x}} \left(\underline{x + \frac{1}{x}}\right) + x + \frac{1}{x}$$

$$= \left[\cancel{\sqrt{3} + \sqrt{2}} + \cancel{\sqrt{3} - \sqrt{2}}\right]^3 - 3 \left[\cancel{\sqrt{3} + \sqrt{2}} + \cancel{\sqrt{3} - \sqrt{2}}\right] + \left[\cancel{\sqrt{3} + \sqrt{2}} + \cancel{\sqrt{3} - \sqrt{2}}\right]$$

$$= (2\sqrt{3})^3 - 3(2\sqrt{3}) + 2\sqrt{3}$$

$$= 24\sqrt{3} - 4\sqrt{3} = 20\sqrt{3} //$$

Q] $\frac{1}{\sqrt{1} + \sqrt{3}} + \frac{1}{\sqrt{3} + \sqrt{5}} + \frac{1}{\sqrt{5} + \sqrt{7}} + \dots + \text{up to } 50 \text{ terms}$

Soln : ① $\frac{\sqrt{1} - \sqrt{3}}{(\sqrt{1} + \sqrt{3})(\sqrt{1} - \sqrt{3})} + \frac{\sqrt{3} - \sqrt{5}}{(\sqrt{3} + \sqrt{5})(\sqrt{3} - \sqrt{5})} +$

$\frac{(\sqrt{5} - \sqrt{7})}{(\sqrt{5} + \sqrt{7})(\sqrt{5} - \sqrt{7})} + \dots + \frac{(\sqrt{99} - \sqrt{101})}{(\sqrt{99} + \sqrt{101})(\sqrt{99} - \sqrt{101})}$

1, 3, 5 - ... upto 50 terms

→ AP

$$\rightarrow T_n = a + (n-1)d$$

$$\rightarrow T_n = 1 + (50-1)2$$

$$= 1 + 98$$

$$= 99$$

$$\textcircled{2} \quad \frac{\sqrt{1}-\sqrt{3}}{(\sqrt{1}+\sqrt{3})(\sqrt{1}-\sqrt{3})} + \frac{\sqrt{3}-\sqrt{5}}{(\sqrt{3}+\sqrt{5})(\sqrt{3}-\sqrt{5})} +$$

$$\frac{(\sqrt{5}-\sqrt{7})}{(\sqrt{5}+\sqrt{7})(\sqrt{5}-\sqrt{7})} + \dots + \frac{(\sqrt{99}-\sqrt{101})}{(\sqrt{99}+\sqrt{101})(\sqrt{99}-\sqrt{101})}$$

$$\left(\frac{\sqrt{1}-\sqrt{3}}{-2} \right) + \left(\frac{\sqrt{3}-\sqrt{5}}{-2} \right) + \left(\frac{\sqrt{5}-\sqrt{7}}{-2} \right) + \dots + \frac{\sqrt{99}-\sqrt{101}}{-2}$$

$$\left(\frac{\sqrt{1} - \sqrt{3}}{-2} \right) + \left(\frac{\sqrt{3} - \sqrt{5}}{-2} \right) + \left(\frac{\sqrt{5} - \sqrt{7}}{-2} \right) + \dots + \frac{\sqrt{99} - \sqrt{101}}{-2}$$

$$-\frac{1}{2} \left[\cancel{\sqrt{1}} + \cancel{\sqrt{3}} + \cancel{\sqrt{3}} - \cancel{\sqrt{5}} + \cancel{\sqrt{5}} - \cancel{\sqrt{7}} + \dots + \cancel{\sqrt{99}} - \sqrt{101} \right]$$

$$= \frac{\sqrt{101} - \sqrt{1}}{2}$$

//

The sum of n terms of the series
 $4+44+444+\dots$ is

(A) $\left(\frac{4}{81}\right)[10^{n+1} - 9n - 1]$ (C) $\left(\frac{4}{81}\right)[10^{n+1} - 9n - 10]$

(B) $\left(\frac{4}{81}\right)[10^{n-1} - 9n - 1]$ (D) $\left(\frac{4}{81}\right)[10^n - 9n - 10]$

[GATE 2011- ECE/EE/IN]

Solⁿ :

4 + 44 + 444 + - - - n terms

m1

$$4 [1 + 11 + 111 + \dots]$$

$$\frac{4}{9} [9 + 99 + 999 + \dots]$$

$$\frac{4}{9} [(10 - 1) + (10^2 - 1) + (10^3 - 1) + \dots]$$

$$\frac{4}{9} \left[(\underline{10}-1) + (\underline{10^2}-1) + (\underline{10^3}-1) + \dots \right]$$

$$\frac{4}{9} \left[(10 + 10^2 + 10^3 + \dots) - \underbrace{(1 + 1 + 1 + \dots)}_n \right]$$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$= \frac{10(10^n - 1)}{10 - 1}$$

$$\frac{4}{9} \left[\frac{10^{n+1} - 10}{10 - 1} - \frac{9n}{9} \right]$$

$$\frac{4}{81} [10^{n+1} - 10 - 9n]$$

The sum of n terms of the series

$4+44+444+\dots$ is

$$\textcircled{n=1} \rightarrow \underline{s_1 = 4}$$

~~(A)~~ $\left(\frac{4}{81}\right)[10^{n+1} - 9n - 1]$

(C) $\left(\frac{4}{81}\right)[10^{n+1} - 9n - 10]$

~~(B)~~ $\left(\frac{4}{81}\right)[10^{n-1} - 9n - 1]$

(D) $\left(\frac{4}{81}\right)[10^n - 9n - 10]$

$\textcircled{M2}$

[GATE 2011- ECE/EE/IN]

[Substitute]

(A) $\frac{4}{81} [10^2 - 9 \times 1 - 1] \neq 4$

(B) $\frac{4}{81} [\cancel{10^{1-1}} - 9 \times 1 - \cancel{1}] \neq 4$

The sum of n terms of the series
 $4+44+444+\dots$ is

(A) $\left(\frac{4}{81}\right)[10^{n+1} - 9n - 1]$ ~~(C) $\left(\frac{4}{81}\right)[10^{n+1} - 9n - 10]$~~

(B) $\left(\frac{4}{81}\right)[10^{n-1} - 9n - 1]$ (D) $\left(\frac{4}{81}\right)[10^n - 9n - 10]$

[GATE 2011- ECE/EE/IN]

~~(C)~~ $\frac{4}{81} [10^{1+1} - 9 \times 1 - 10] = \frac{4}{81} \times 81 = 4$

(D) $\frac{4}{81} [10 - 9 - 10] \neq 4$

If $q^{-a} = \frac{1}{r}$ and $r^{-b} = \frac{1}{s}$ and $s^{-c} = \frac{1}{q}$,

The value of abc is

(A) $(rqs)^{-1}$

~~(B) 0~~

(C) 1

~~(D) $r+q+s$~~

[GATE 2016- ECE/ME]

$$b = \frac{\ln s}{\ln r}$$

$$c = \frac{\ln q}{\ln s}$$

M1

$$q^{-a} = \frac{1}{r}$$

$$\Rightarrow \ln [q^{-a}] = \ln (r^{-1})$$

$$\therefore a \ln q = -1 \ln r$$

$$a = \ln r / \ln q$$

. a b c

$$= \frac{\cancel{\ln r}}{\cancel{\ln q}} \times \frac{\cancel{\ln s}}{\cancel{\ln r}} \times \frac{\cancel{\ln q}}{\cancel{\ln s}}$$

$$= 1$$

If $q^{-a} = \frac{1}{r}$ and $r^{-b} = \frac{1}{s}$ and $s^{-c} = \frac{1}{q}$,

The value of abc is

(A) $(rqs)^{-1}$

(B) 0

☒ (C) 1 ☒

(D) $r+q+s$

[GATE 2016- ECE/ME]

M2

Assume

$$r = q = s = k$$

$$k^{-a} = \frac{1}{k} = k^{-1}$$

If $q^{-a} = \frac{1}{r}$ and $r^{-b} = \frac{1}{s}$ and $s^{-c} = \frac{1}{q}$,

• The value of abc is

(A) $(rqs)^{-1}$

(B) 0

(C) 1

(D) $r+q+s$

[GATE 2016- ECE/ME]

$$q = s^c = (r^b)^c = r^{bc}$$

$$= (q^a)^{bc}$$

$$q^1 = q^{abc}$$

M3

$$\Rightarrow q^{-a} = \frac{1}{r}$$

$$\Rightarrow r = q^a$$

$$\Rightarrow s = r^b$$

$$\Rightarrow q = s^c$$

$$q^{-a} = \frac{1}{r}$$

$$\frac{1}{q^a} = \frac{1}{r}$$

$$r = q^a$$